

Gabriele Novello

Graphics programmer with experience in Vulkan and real-time rendering systems.

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EDUCATION

University of Turin 2024 – 2026
M.Sc., Computer Science Turin, Italy

- part-time position: translated computer graphics course slides from Italian to English
- master's thesis: a probe-based framework for dynamic real-time global illumination
- major topics studied: computer graphics, virtual reality systems, image processing & computer vision, parallel & distributed systems, machine learning & deep learning

University of Turin 2021 – 2024
B.Sc., Computer Science Turin, Italy

- final grade: 110/110, cum laude
- bachelor's thesis: game state saving approaches in gaming
- developed a small roguelite game in Godot to test the methods proposed in the thesis

PERSONAL PROJECTS

Lattice-Boltzmann fluid solver 2026
Parallelization and performance analysis

- parallelized C implementation with OpenMP and OpenMPI
- analyzed scalability on a cluster of 36 nodes using Slurm as a workload manager

Felina 2025 – Present
Real-time Vulkan Renderer

- built with C++, modern Vulkan (1.3+), CMake and HLSL
- features PBR materials, deferred shading and multiple punctual lights support
- includes Dear ImGui user-interface and scene loading through the glTF file format

Monte Carlo Denoiser 2024 – 2025
Standalone Tool / Research

- developed and trained a ResNet-based neural network in Python using PyTorch to denoise Monte Carlo renderings using Disney's published dataset and methodology

TECHNICAL SKILLS

Programming & APIs

C/C++ Python Java
Vulkan Slang HLSL
CUDA OpenMP MPI

Tools

Git RenderDoc CMake
Visual Studio Godot Unreal Engine

LANGUAGES

- English · proficient
- Italian · native

INTERESTS

- cinematography, filmmaking and storytelling
- digital and film photography
- video games and game development
- classic and electric guitar, music production
- learning: computer science, physics and foreign languages